

Education

University of Waterloo

Ontario, Canada

MASe in Mechanical and Mechatronics Engineering | GPA: 88.5

Sep. 2024 – Present

Advisor: Prof. Soo Jeon

Relevant Coursework: Optimal Control, Adaptive Control, Video Processing, Machine Learning

The Hong Kong Polytechnic University (HKPolyU)

Kowloon, Hong Kong

B.Eng. (Honours) in Mechanical Engineering

Sep. 2018 – Jun. 2024

Graduated with First Class Honors

(Mandatory military service: 2021–2023)

Thesis: *A Mobile Robot with an Active Suspension System for Navigation in Uneven Terrain*

Publications

T.H. Kweon, S. Jeon. “Semantic-Aware Active Perception for Next-Best-View Grasp Planning.” *International Journal of Precision Engineering and Manufacturing (IJPEM)*, In Press (Accepted), 2026. [Preprint]

Research & Work Experience

Mechanical Systems Control Lab, UWaterloo

Ontario, Canada

Graduate Student Researcher

Sep. 2024 – Present

Advisor: Prof. Soo Jeon

- Developed a semantic Next-Best-View framework for grasp planning under occlusion, integrating semantic segmentation, volumetric mapping, and grasp synthesis.
- Developed a VLM-distilled policy that predicts the Next-Best-View for grasping via a lightweight network, achieving $2.6\times$ faster inference with comparable success to information gain baselines.

Robotics and Machine Intelligence Lab, HKPolyU

Kowloon, Hong Kong

Undergraduate Research Assistant

Dec. 2020 – Jun. 2021

Advisor: Prof. David Navarro-Alarcon

- Built a mobile robot capable of autonomous navigation over liquid and muddy terrain.
- Implemented a vision-based localization and navigation system using artificial markers.
- Contributed to securing project funding.

Origami Labs (Start-up)

Tsuen Wan, Hong Kong

Engineering Intern

Jul. 2020 – Sep. 2020

- Integrated an accelerometer-based trigger system for automatic microphone activation.
- Applied signal-processing filters to reduce environmental noise in wearable prototypes.

Selected Projects

VLA Policy Evaluation System

Feb. 2026 – Apr. 2026

Personal Project

- Collected robot manipulation trajectory data via Meta controller-based teleoperation and built a training pipeline.
- Fine-tuned a $\pi_{0.5}$ -based VLA model on the collected demonstrations.
- Implemented a grasp success evaluation pipeline using rule-based prompting and Qwen2.5-VL.

End-to-End Visual Grasping with PPO, UWaterloo

Jan. 2025 – Apr. 2025

SYDE 673 Video Processing and Analysis (Course Project)

- Developed and trained a PPO-based visuomotor grasping policy using the Barrett WAM arm in the Genesis simulator.

- Designed a visual reward function based on corner overlap, contact detection, and lift height.
- Built a multi-environment camera system for robust, parallelized vision-based training.

Adaptive Control of a 2-DOF Robotic Arm, UWaterloo Sep. 2024 – Dec. 2024
ME780 Adaptive Control (Course Project)

- Implemented a Model Reference Adaptive Control (MRAC) scheme to adaptively estimate unknown dynamics and ensure stable trajectory tracking.
- Derived Lyapunov-based adaptation laws and validated controller performance in MATLAB/Simulink.

Active Suspension for Navigating Rough Terrain, HKPolyU Sep. 2023 – Jun. 2024
Undergraduate Thesis

- Designed a mobile robot with four independently actuated suspension units driven by stepper motors.
- Integrated IMU feedback to detect terrain pitch and actively adjust body height for stable navigation improving robustness on uneven terrain.

Teaching Experience & Service

Peer Reviewer 2026

- International Journal of Precision Engineering and Manufacturing (IJPEM)
- IEEE ICRA Workshop (Act to Sense to Act Better)

ME549/MTE544 Autonomous Mobile Robotics, University of Waterloo
Teaching Assistant Winter 2025, Fall 2025, Winter 2026

- Led lab sessions on localization, control, and planning; held office hours; graded assignments and exams.

Scholarships & Awards

International Master's Award of Excellence (IMAE), UWaterloo 2024 – 2026
Dean's Honour List, HKPolyU 2021
Full Entry Scholarship, HKPolyU 2018 – 2024

Extracurricular Activities

Peer Tutor in AMA1120 (Basic Mathematics II), HKPolyU Jan. 2024 – May 2024
Mentor, Korean Student Association, HKPolyU Sep. 2023 – Dec. 2023
Team Leader, STEM Learning Kits for Overseas Students, HKPolyU Jun. 2020 – Aug. 2020

Skills

Programming: Python, C++, MATLAB, Shell
Tools: PyTorch, ROS 1 & 2, OpenCV, Open3D, Git, Docker, L^AT_EX
Robots: Franka Panda, Barrett WAM, TIAGO++, TurtleBot
Simulation: PyBullet, Genesis, Gazebo, Simulink
Hardware: IMU, RealSense L515/D455, LiDAR, Arduino, Jetson Orin, Raspberry Pi